

ROLL-OFF CONTAINER — MARKET COMPARISON AND EXPORT STRATEGY

TR Domestic Market · TR→US Export · TR→EU Export · Manufacturing in the US

Focus: Remaining competitive in the US and EU markets without compromising quality

Date: August 2026 · **Currency:** All values are in US Dollars (\$) unless otherwise stated; European market prices are additionally shown in €. Turkish domestic values are TRY-based; USD equivalents are given at 1 USD ≈ ₺47.5.

1. EXECUTIVE SUMMARY — DECISION TABLE

Criterion	A) TR domestic	B) TR→US export	C) TR→EU export	D) US plant (Georgia)
Incremental investment	\$1.78M (full plant)	~\$0.3–0.5M (standardization + certification + sales office)	~\$0.2–0.4M (EN 1090 + low-carbon steel)	\$3.37M (full plant)
Unit cost	~\$6,674 → \$6,442	~\$6,674 → \$6,442	~\$6,674 → \$6,442	~\$12,354 → \$11,050
Selling price	~\$10,526–11,474	FOB ~\$9,000 → landed ~\$11,800	FOB ~\$8,000 → landed ~\$9,850	\$14,500–15,800
Market price	~\$11,000	\$14,500	€8,000–9,000 (~\$9,300–10,400)	\$14,500
Margin	38% → 46%	~35% (FOB)	~20% (premium end)	17% → 34%
Payback	~15 months	~2 years	~3 years	~4 years
Risks	FX, political, dependence on customs protection	Section 232 (25% tariff), Buy America, warranty/service	CBAM, Polish price competition, low price ceiling	Price competition incl. imports, investment intensity
Priority	1 (base)	2 (conditional — if 232 clarified)	3 (premium niche complement)	4 (strategic/local preference if any)

Key messages:

- **Highest return is in the TR domestic market** (protection + low cost + 38–46% margin).
- **US export offers the highest price arbitrage** (landed ~\$11,800 vs market \$14,500) — but the single breaking point is Section 232.
- **Europe is geographically close but price-pressured** — profitable only at the premium/custom end and with EN 1090 + low-carbon steel.
- **A US plant is more expensive than importing from Turkey** — it makes sense only in Buy America / local-preference channels.
- **Recommended hybrid:** production in TR + light assembly (KD) and a sales line in the US → quality proven locally, tariff risk reduced, well below the plant investment (~\$3.37M).

2. FOUR OPTIONS — FINANCIAL COMPARISON

2.1 Unit economics (same product, Phase 1 cost basis)

(\$/unit)	TR domestic	TR→US	TR→EU	US plant
Production cost	\$6,674	\$6,674	\$6,674	\$12,354
FOB / sale price	\$10,526	\$9,000	\$8,000	\$14,500
Freight	—	\$2,200	\$1,200	—
Customs + tax	—	~\$270 (3%) / +\$2,250 (232 risk)	~\$650 (CBAM)	—
Landed cost	\$10,526	~\$11,800	~\$9,850	\$14,500
Market price	~\$11,000	\$14,500	€8,000–9,000 (~\$9,300–10,400)	\$14,500
Exporter/plant margin	~38%	~35%	~20%	~17%

Note: The TR domestic selling price is the USD equivalent of ₺500,000 (1 USD ≈ ₺47.5). In the US and EU columns, values are the exporter's USD receipts; only the EU market price is EUR-based.

The US column is critical: **landed from Turkey (~\$11,800) is even below the Georgia plant's production cost (~\$12,354)**. On pure price, exporting to the US is cheaper than manufacturing in the US.

2.2 Investment and payback

	TR domestic	TR→US	TR→EU	US plant
Investment (\$M)	\$1.78M	+\$0.4M (incremental)	+\$0.3M (incremental)	\$3.37M
3-yr cumulative FCF (\$M)	+\$3.35M (1.9×)	—	—	+\$1.18M (0.35×)
Simple payback	~15 months	~2 years	~3 years	~4 years

3. US STRATEGY — COMPETITIVENESS WITHOUT COMPROMISING QUALITY

3.1 Product standardization (market fit = quality, not waste)

Action	Effect	Quality impact
Width 2,500 → 2,440 mm (8 ft)	Fits a standard 40 ft container/flat rack → freight/unit drops ~15–20%	None (8 ft is already the US standard)
US hooklift compliance (Galbreath/Perkins hook, 18–20 ft skid)	Direct fleet compatibility → lower rejection/adaptation risk	Increases (full spec compliance)
Position as 30 m³ → 40 yd³ (~30.6 m³)	US buyers classify the product correctly	None
Nestable (nested stacking) body design	2–3 levels on vessel → freight/unit drops ~40–50%	Increases (FEM reinforcement; sloped wall eases emptying)

3.2 Logistics and freight optimization

- Regular service agreement with the **Port of Savannah/Brunswick**; for high volumes, replace RoRo/breakbulk with **container + nestable packing** → freight per unit can drop from \$2,200 to ~\$1,400.
- **Offer FOB instead of DDP**: freight and FX risk shared with the distributor.
- **Shipment consolidation**: full loads with 4–8 week batched orders; group shipments by customer.

3.3 Section 232 tariff strategy (the clause that decides the project)

Option	How it works	Advantage
Correct HTS classification	Product-specific customs counsel opinion (7326.90 vs 7309 vs 8716); some manufactured steel articles are out of scope	~3% duty instead of 25%
Product-specific exclusion (PSE)	Exclusion petition to the US Department of Commerce	Legal 25% exemption
KD (knock-down) assembly in the US	Steel parts made in TR, welding/assembly on a small US line (~\$0.5–0.8M) → "Made in USA"	Tariff limits + approach to Buy America criteria
Mexico assembly (USMCA)	TR parts → MX assembly → zero-duty US entry (if USMCA rules met)	0% duty

Recommendation: obtain an HTS opinion and file the PSE before the first pilot shipment; if the 25% tariff stays, switch to the KD assembly scenario.

3.4 Quality perception and certification (breaking the "cheap Turkish goods" perception)

- **AWS D1.1**: All welders and procedures certified to the American welding standard → a direct answer to US customer specifications.
- **ISO 9001 + AISC/manufacturer quality program**: factory audit reports added to tender documents.
- **Warranty line in the US**: on-site warranty through a distributor/service network — a "quality + service" package (reduces quality recall risk).
- **Test/report package**: load test (FEM + physical), weld inspection, coating thickness — delivered with every shipment.

3.5 Market entry (low sunk-risk)

- Production in TR + **small warehouse/sales office in Georgia** (not a plant) → stocked fast delivery and on-site contact.
- **Channels**: hooklift dealers, solid waste companies (WM, Republic, Waste Pro regional), rental companies; a "Made in USA" KD variant for public tenders.

4. EU STRATEGY — COMPETITIVENESS WITHOUT COMPROMISING QUALITY

4.1 Certification (quality assurance = entry condition)

- **EN 1090-1 (EXC2/EXC3)**: CE marking for structural steel products — though not always mandatory, it is a prerequisite for German/Austrian buyers. Strong quality signal.
- **TÜV/GS inspection** and **EN ISO 9606**-certified welders.

- **ISO 9001** + technical file with every product (welding plan, material certificates, test results).

4.2 CBAM (Carbon Border Adjustment Mechanism) management — full application from 2026

Action	Effect
Low-carbon steel sourcing (EAF/scrap-based, low embedded CO ₂)	CBAM certificate cost drops from ~\$650 to ~\$300 (CBAM is EUR-denominated; values shown in USD equivalent)
CBAM reporting infrastructure (embedded emissions documentation)	Correct tax base; zero penalty risk
Track Turkey's ETS (entry into force 2027+)	Burden decreases gradually as the domestic carbon price is deducted from CBAM
Long-life coating (HDG/2K paint option)	Quality rises, warranty/customer satisfaction → pushes the price ceiling up

4.3 Product adaptation (Europe = quality standard, not waste)

- **German/Austrian hook systems** (Abrollkipper) and Swiss/French variants — 2.5 m width is already standard in Europe (truck 2.55 m) → **no logistics or fit issue**.
- **Special variants:** drain-valve, fire/concrete demolition container, compactor-compatible, stainless-lined → €8,000–11,000 premium band.
- **Short lead time (6–8 weeks):** China 12+ weeks, near-Polish levels — lead time justifies the premium price.

4.4 Positioning and channels

- **Do not enter the standard market** (€7,000 band; Polish competition). **Compete at the premium/custom end** (€8,000–11,000).
- **Channels:** Abfallwirtschaft companies (Germany/Austria), municipal contracts, rental fleets, OEM agreements with hooklift manufacturers.
- **Logistics:** 2 units per truck, Izmir/Trieste RoRo, group shipments — freight ~\$1,200/unit.

5. COMMON PRINCIPLE — COST REDUCTION WITHOUT COMPROMISING QUALITY

These principles apply to both US and EU exports; they reduce cost **without reducing material thickness** (6 mm floor and 4 mm wall are preserved).

Principle	How	Gain
FEM design optimization	Reduce weight 5–10% (same structural strength; added reinforcement points)	Lower material + freight, higher quality
Nestable/stackable body	2–3 levels on vessel	Freight ~40–50% lower
Phase 3 automation (robots + CNC plasma)	Consistent welding, low rework	Unit cost ~6% lower, quality standardized
Few variants, high series	Jig/robot efficiency; single loading configuration	Efficiency +15–20%
Bulk steel + scrap hedge	Long-term supply contract	Lower price risk

Principle	How	Gain
Long-life coating (HDG/2K)	Lower warranty and service cost	Lower total cost of ownership (premium defense)
Certification set (AWS D1.1 / EN 1090 / ISO 9001)	Opens tender and distributor doors	Margin protected, no need to cut price

6. RECOMMENDED ROADMAP (HYBRID)

Step	Timing	Action
1	Months 0–3	Phase 1 production at the TR plant + obtain HTS classification and Section 232 opinion (counsel)
2	Months 3–6	Standardize product 2,440 mm + nestable + US/EU hook compliance ; start AWS D1.1 / EN 1090 certification
3	Months 6–12	Pilot shipments: 40–60 units to the US (with distributor), 20–40 premium/custom units to the EU
4	Months 12–18	Decide on real sales data: if 232 applies, KD assembly in the US (~\$0.5–0.8M); otherwise scale direct exports
5	Months 18–36	Trigger Phase 3 robotic investment with export volume; run the EU premium line and US channels in parallel

Decision logic: Export decisions rest on **real pilot shipment data**; no large sunk investment is made in any market in advance. A US plant (\$3.37M) only comes up if Buy America / local-preference channels are won.

7. DISCLAIMER

All figures in this document are **estimates** based on publicly available data as of August 2026 (freight, tariffs, CBAM prices, steel). Section 232 scope, CBAM certificate price and HTS classification require official opinions and customs counsel review. This document is not investment advice.